

Applied science faculty

CHIM0699-2 : Life cycle assessment - Ecodesign

LCA Review on Spanish Cement Industry

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1 Introduction

On average, approximately 1 ton of concrete is produced each year for every human being in the world. The cement industry is essential in terms of infrastructure but it represents an environmental challenge due to its intensive energy consumption and emissions. In 2010, the global production of cement was 3,310Mt and 191 Mt in the European Union, from which 23.5Mt was produced in Spain.[1]

These 23.5 Mt might seem a fraction of the global output, but it represented approximately 6.4% of Spain's total CO_2 emissions for that year, which is a considerable part of these emissions. Therefore, an LCA is required to better understand the contribution of cement production to these impacts.

First, the clinker production process is analyzed using 1 ton of clinker as the functional unit. Then, different cement types are compared in order to evaluate the influence of clinker content and supplementary materials on environmental performance. Finally, an alternative fuel scenario is assessed and compared to the base scenario to explore potential improvements in the cement production system.

The document will include the modeling of the clinker and cements, a comparison between the different cements, and also an analysis contrasting the base scenario with a more sustainable alternative. The comparison aims to highlight the environmental benefits of adopting sustainable practices and to identify the key processes and inputs that affect most significantly to the environmental impacts.

2 Critical review of the reference report

The report by García-Gusano et al. (2015) [2], titled *"Life cycle assessment of the Spanish cement industry: implementation of environmental-friendly solutions"*, it is the reference document selected for this LCA study and was reviewed to verify if it is aligned with the methodology defined in ISO 14040/44.

2.1 Article accordance with the ISO standards

The article strictly follows the four steps required by the ISO standard: Goal and Scope definition, Inventory Analysis, Impact Assessment, and Interpretation.The information is clear and complete and the structure is logical.

2.2 Goal and scope

The objective of the study is clear: to identify the main environmental hotpots of the Spanish cement industry (using 2010 data) and to evaluate potential improvements through different type of alternative scenarios. They follow a cradle-to-gate approach, covering raw material extraction, transport, and clinker/cement production.

2.3 LCA steps

The LCA is transparent in terms of inventory, with detailed tables of inputs, fuels, emissions... which are very useful to introduce the input data in SimaPro. But, the assessment does not cover all impact categories. Important ones like, for example, Fossil Resource Scarcity, Land Use, Water Use, and Particulate Matter formation are not included. So the analysis is relatively complete, but still misses some relevant dimensions. Also, a small error was found in the clinker inventory: the input for the conveyor belt has an incorrect magnitude.

2.4 Data Collection and Validation

The main data in the report comes from trusly sources like OFICEMEN (2010a/b), which is the Spanish Cement Manufacturers Association. The text continually makes references to trusted sources such as the BOA (Spanish law), ISO sources, IEA(International Energy Agency) and some background data was supplemented by Ecoinvent v2, which is also reliable source.

The main limitation could be the age of the data, since the sector has evolved in terms of thermal efficiency and alternative fuel usage.Therefore, while this dataset is perfectly suitable for the academic purpose of this project, its validity for a realistic assessment of the current Spanish cement industry is limited, since the information may not accurately reflect modern efficiency standards

3 Goal and scope

The purpose of this report is to get familiar with the SimaPro software by modeling an existing Life Cycle Assessment (LCA) and conducting a critical review of the article [2]. The article analyzes the Spanish cement sector in 2010 with LCA and evaluates scenarios where best available technologies and substitution measures are taken into consideration.

This LCA is a cradle-to-gate analysis using a cut-off approach, meaning that no benefits are allocated to the use of "waste", i.e., waste is considered "free", and only the recycling processes and transport are accounted for according to the polluter-pays principle. The system boundaries extend from extractive processes to cement production, as shown in Figure 1.

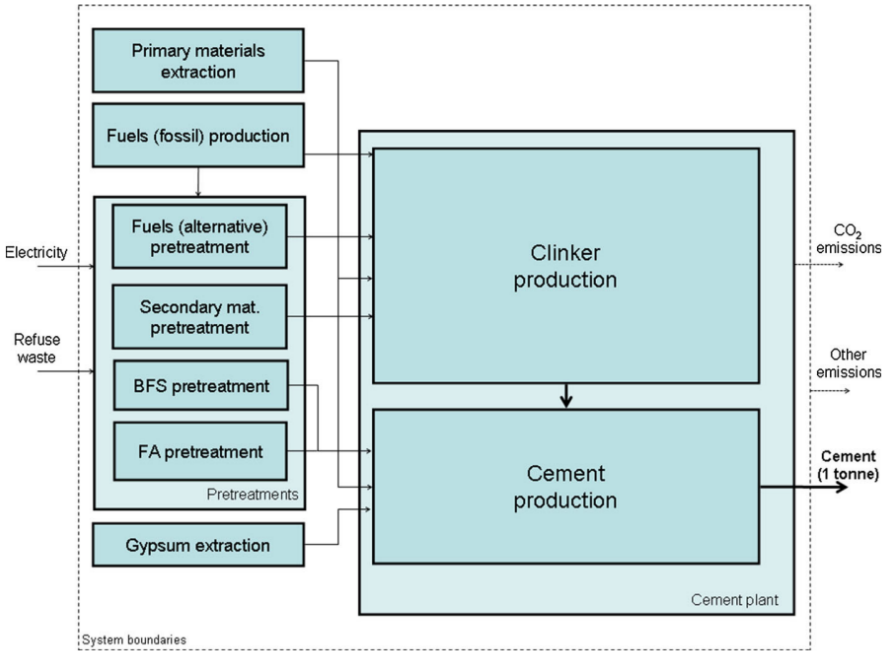


Figure 1: Scheme of the main process in cement production in this LCA study

The work is divided into two analyses:

1. First, the analysis of 1 ton of clinker as a functional unit. As clinker is the same for each cement type, it is interesting to study the impact of this production line.
2. Second, the analysis of different cements is carried out using 1 ton of cement as the functional unit. Only five cements are analyzed among all the cements presented in the reference article. The cement names and compositions used in this work are shown in Table 1.

Cement Type	Clinker (%)	BFS (%)	PZ (%)	FA (%)	Limestone (%)
CEM I (Portland)	97.5	-	-	-	-
CEM II/A-L (Portland II)	87.0	-	-	-	13.0
CEM III/A (Blast furnace)	64.0	36.0	-	-	-
CEM IV/A (Pozzolanic)	85.0	3.8	11.3	-	-
CEM V/A (Composite)	52.0	27.8	10.0	10.0	-

FS: Blast Furnace Slag; PZ: Pozzolana; FA: Fly Ashes. The remaining percentage corresponds to gypsum.

Table 1: Cement composition.

The method used is the same as the article, namely, the ILCD 2011 midpoint method. The following impact categories have been analyzed: *Climate change* (CC), *human toxicity with cancer effect* (HTce), *photochemical ozone formation* (POF), *acidification* (A), and *freshwater eutrophication* (FE).

The software SimaPro 10.2.0.3 has been used to build the LCA, and all processes have been taken from the Ecoinvent 3 database.

4 Modeling

4.1 Electricity mix

The first thing that was modeled was the electricity mix. It's an essential process used throughout the whole process. The model was copied from an existing 'Electricity mix high-voltage ES', and the inputs from technosphere were adapted to match as closely as possible the inputs from the Article shown in Table 8.

Table 2 shows the new electricity mix model. The major changes are the division of the nuclear and wind input. Indeed, the existing electricity mix in Spain includes nuclear reactors that produce both steam and boiling water. It was decided that the nuclear input from the reference article shown in Figure 9 would be divided proportionally based on the proportion of the existing 'Electricity mix high-voltage ES', 80% for the pressure and 20% for the boiling water reactor. The same reasoning was made for the wind electricity input, which will be divided into 2 categories from the existing electricity mix from Spain: onshore <1 MW turbine, and 1-3 MW turbine. The proportion was 64% for the 1-3 MW onshore turbine 36% for the <2 MW onshore turbine.
Additionally, since no gas oil, fuel, propane cogeneration is found in the database, this electricity input process was simply added to natural gas cogeneration. Finally, the municipal waste input was modeled from an existing medium voltage electricity mix that includes an existing municipal waste process.

Technology	**kWh (input)**
Hard coal	0.0810
Lignite	0.0040
Natural gas, NGCC	0.2308
Natural gas, Gas Turbine	0.0065
Hydropower, large	0.1381
Hydropower, run-of-river	0.0244
Nuclear, pressure water reactor	0.1773
Nuclear, boiling water reactor	0.0442
Solar, photovoltaic	0.0219
Solar, concentrated solar power	0.0025
Wind onshore, 1-3 MW	0.0996
Wind onshore, <1MW	0.0548
Biomass	0.0088
Biogas generation	0.0025
Natural gas cogeneration	0.0984
Municipal solid waste	0.0052
Electricity from the mix	**kWh (output)**
Electricity	1

NGCC natural gas combined cycle plant.

Table 2: Electricity mix

4.2 Clinker production

The LCA model for the production of 1 ton of clinker integrates different types of input data, including primary materials, secondary materials, infrastructure, transport, fossil fuels, alternative fuels, electricity, and emissions.

During the inventory modeling, inputs are mainly based on an existing Swiss (CH) clinker production process, as it represents the most accurate aproximatio for Spanish cement production standards. In addition, processes geographically closest to Spain are preferred to improve the representativeness of the model. Market processes are also favored, since the system is modeled as a gate-to-gate process. This step is important to complete the inventory and to match, as closely as possible, the processes described in the reference article.

Most of the processes model follow the reference report, namely: Primary materials, infrastructure, and fossil fuels. However, some assumptions and issues were identified in the remaining inputs during the LCA development, as shown in the following sections.

Secondary materials

The secondary materials were modeled by summing up the secondary materials list in Table 9, except blast furnace slag, in a new process called 'Grouped secondary materials'. This process was copied from an existing 'Blast furnace slag' process, but without the transport from RoW and BR, by leaving only the transport from RER. The new secondary materials process can be summarized in Table 3.

Secondary materials	Amount
Grouped secondary materials (t)	2.31E-3
Blast furnace slag (t)	3,16E-3

Table 3: List of secondary materials

Transport

For the modeling of the **'Lorry 20-28 t (tkm)'** and **'Lorry >28 t (tkm)'**, no process with these exact distances can be found in the ecoinvent database. Instead, lorry 16-32 metric tons and >32 metric tons were used, respectively. Another specification is that the EURO 5 standard was used. Indeed, this was the most appropriate choice since the process studied is the cement industry in Spain in 2010, and the Euro 5 emissions standard came into effect in 2009 [3].

Alternative fuel

For the alternative fuels category, 'used tyres' are selected as the single representative process. According to the inventory data for the Spanish cement industry in Table 9 , used tyres provide

the most significant energy contribution among alternative fuels with a value of 0.172 GJ/t. This value is higher than those of other alternatives, such as municipal sewage sludge or plastics, which contribute 0.03 GJ/t and 0.025 GJ/t, respectively. This simplification allows the energy and environmental impacts to be assessed without modelling each minor fuel fraction individually.

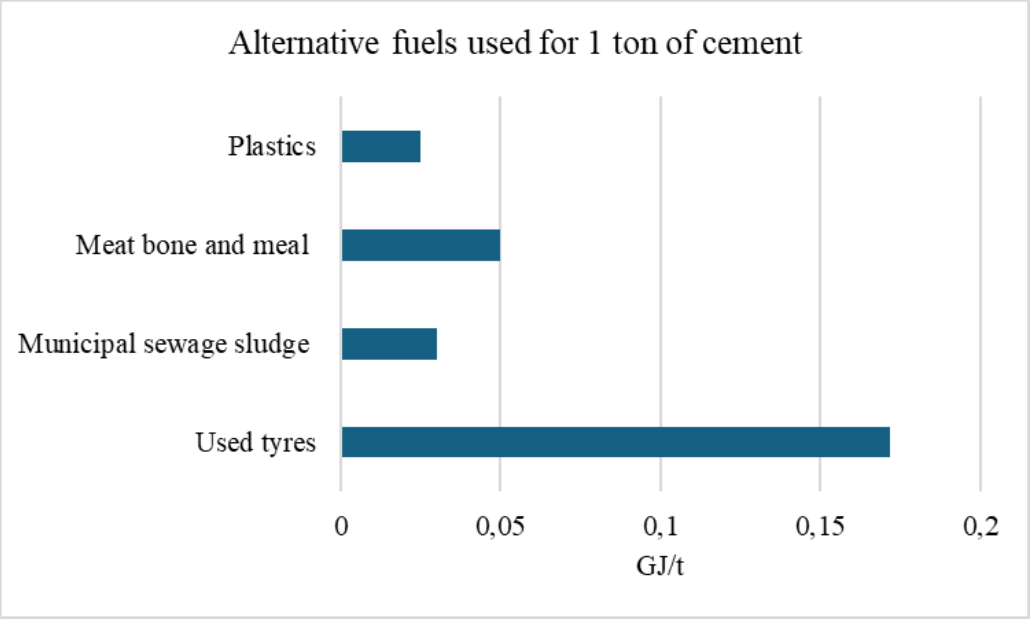


Figure 2: Alternative fuels used per ton of cement, with used tyres representing the dominant share

Emission

The emission of the clinker production was modeled in two parts. One directly from the clinker production process, since it is a direct emission from the clinker production. The other are linked to the combustion of fossil fuels. To take the latter emissions into account, a new process was created that models only the emissions from the fossil fuels called 'Hard coal emission', 'Heavy fuel oil emission', and 'Petcoke emission'. It was then implemented into the clinker process by keeping the same amount of fossil fuel as the inventory in Table 9. The emissions related to fossil fuels are shown in Table 10.

Moreover, the relevant air emissions through the 'Used Tyres' process were directly introduced as outputs of this process. Table 4 shows the emissions of used tyres.

Item	Unit	Amount
Input		
Low hearting value	MJ/kg	31.34
Transport, lorry >28t (2)	tkm	0.05
Electricity, medium voltage (2)	kWh	0.045
Emissions to air		
Carbon monoxide, fossil	kg	2.23E-04
Carbon dioxide, fossil	kg	3.09E+00
Methane, fossil	kg	6.38E-06
Sulphur dioxide	kg	1.58E-05
Cadmium	kg	4.41E-10
Mercury	kg	3.45E-17
Lead	kg	2.60E-09
Zinc	kg	2.61E-07
Emissions to water		
Chemical Organic Demand river	kg	5.70E-05
Chemical Organic Demand groundwater	kg	2.11E-02

Table 4: Input and Emission Data for 'Used Tyres'

Additional Modeling Considerations

There are some details to consider when developing the model that can cause the entire LCA process to be inaccurate if they are not properly fixed.

For example, when transport inputs are introduced into the SimaPro model following the reference report inputs in Table 9, a conveyor belt distance of 2 km is specified. However, a 2 km conveyor belt distance resulted in disproportionate impact results. This indicates that the value reported in the reference study is not realistic and that an appropriate order of magnitude for the conveyor belt process must be found.

Consequently, a value of 2×10^{-8} meters for the conveyor belt is used in the LCA. This value was derived from other clinker processes by using the "used by" function in SimaPro, and then a more realistic contribution to the overall results is obtained.

Care must also be taken when analyzing the results, because SimaPro calculates them by default for a 1 kg reference unit. This must be manually changed to a 1 ton reference; otherwise, the results will be off by a factor of 1000.

4.3 Cement production

Once the clinker production process is completed, the material inputs for producing 1 ton of cement vary depending on the type of cement required, as shown in the Table 1. It is important to note that the remaining percentage in the composition balance corresponds to gypsum, as explained in the reference study [2].

The composite cement uses fly ash; however, it is not considered in the analysis due to the application of a cut-off approach. Additionally, electricity and particulate matter, PM $_{10}$, are added to every type of cement, and are shown in Table 5. The electricity process used is the same as for the clinker electricity model in Table 2.

Item	**Amount**
Input	
Materials	
Production breakdown by cement subtype (Table 1)	–
Electricity	
Electricity (MWh)	3.91E-02
Output	
Emissions	
PM_{10} (t)	4.57E-06

Table 5: General Cement Production Input/Output Data

5 Environmental impact results

5.1 Clinker production

The indicators can be compared to those of the article. Table 6 shows the results obtained by both studies.

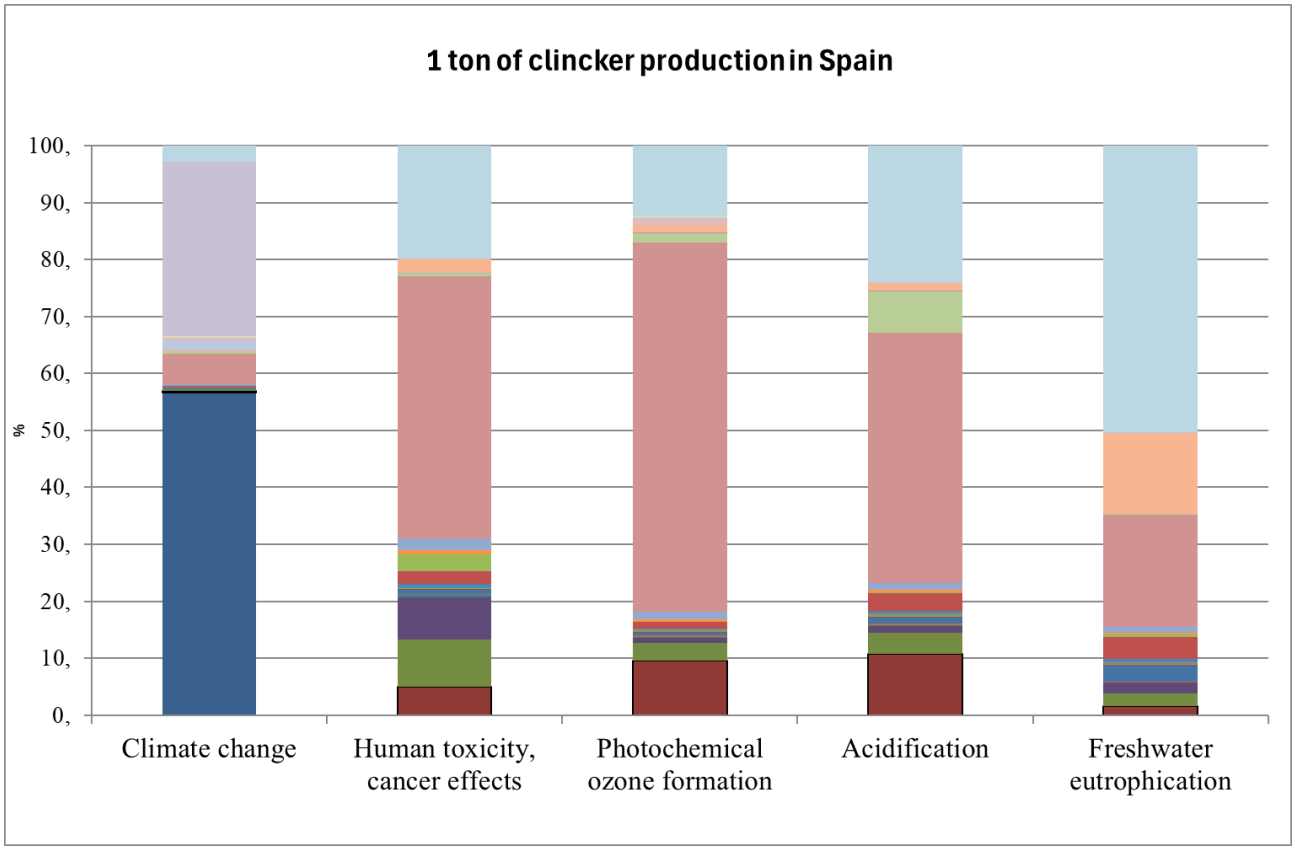
Impact category	**1 ton of clinker (García results)**	**1 ton of clinker (Current study's results)**
Climate Change (kg CO_2 eq)	$9.29E+02$	$9.32E+02$
Human toxicity with cancer effects (CTUh)	$1.20E-06$	$2.32E-05$
Photochemical Ozone Formation (kg NMVOC eq)	$1.24E+00$	$7.18E-01$
Acidification (molc H^+ eq)	$3.93E+00$	$5.65E-01$
Freshwater Eutrophication kg P eq)	$1.21E-02$	$1.48E-02$

Table 6: Characterization Analysis for Clinker and Cement Production

The results obtained are very close to those reported by García for the climate change category. For human toxicity with cancer effects, a higher value is observed, while lower values are obtained for photochemical ozone formation and acidification. Freshwater eutrophication results are also very similar, just slightly higher. These differences can be attributed to modifications made to the original study and to updates in the Ecoinvent 3 database.

A comparison between the two characterization graphs can be carried out. Figure 3 presents the characterization graph obtained and can be compared with the results from García shown in Figure 8. In the reference article, the results are regrouped by production stages. However, since no detailed information on the processes included in each production stage is provided, a similar regrouping cannot be performed in the present study.

An analysis of the characterization graph shown in Figure 3 can be carried out. The climate change impact is divided into two main contributions, namely clinker production and petcoke emissions. Human toxicity, photochemical ozone formation, and acidification are split into three main contributions: electricity production, petroleum combustion, and raw material mining, i.e., limestone, clay, and calcareous marl. Fossil fuel combustion has a greater impact than the other two processes. For freshwater eutrophication, electricity production is the main contributor, accounting for about 50%, while fossil fuel combustion contributes about 30% of this impact category. In summary, the energy and fuel consumption phases are the most significant stages in the clinker production process.



- Electricity, high voltage {ES}| cement LCA
- Petcoke emission
- Hard coal emission
- Used Tyres
- Hard coal briquettes {RER}| market for hard coal briquettes | Cut-off, U
- Heavy fuel oil, burned in refinery furnace {GLO}| market for heavy fuel oil, burned in refinery furnace | Cut-off, U
- Coke {GLO}| petroleum coke to generic market for coke | Cut-off, U
- Industrial machine, heavy, unspecified {RER}| market for industrial machine, heavy, unspecified | Cut-off, U
- Grouped secondary materials
- Ground granulated blast furnace slag {RoW}| market for ground granulated blast furnace slag | Cut-off, U
- Kaolin {GLO}| market for kaolin | Cut-off, U
- Calcareous marl {RER}| market for calcareous marl | Cut-off, U
- Limestone, crushed, for mill {CH}| market for limestone, crushed, for mill | Cut-off, U
- 1 ton clinker

Figure 3: LCIA results of the production of 1 t of clinker in 2010 by production stage

These contributing elements can be examined in more detail. For the climate change category, the combustion of fossil fuels releases carbon dioxide, which contributes to global warming. In addition, the chemistry of clinker production includes the calcination of limestone, which inherently generates CO_2, as shown in Equation 1.

$$CaCO_3 \rightarrow CaO + CO_2 \tag{1}$$

For human toxicity with cancer effects, the Comparative Toxic Unit for humans (CTUh) is assessed using the USEtox model. This model estimates the final impact based on the environmental fate of chemicals, human exposure pathways, and the probability of cancer occurrence once the substances enter the human body. It mainly focuses on two major groups: heavy metals and organic compounds [4]. The main contributors to the clinker production process are fossil fuel combustion, electricity production, and mining activities. Petroleum coke combustion releases heavy metals such as nickel, vanadium, and selenium, as well as polycyclic aromatic hydrocarbons [5]. Impacts related to electricity are largely upstream, originating from the extraction of raw materials such as copper and zinc. Impacts from mining mainly involve arsenic and chromium, which are naturally present in the earth and can be released as dust or leached into groundwater during extraction.

Photochemical ozone formation and acidification show similar patterns and share the same main contributors, i.e., fossil fuel combustion, electricity production, and mining, and are therefore analyzed together. Photochemical ozone formation is characterized by the emission of non-methane volatile organic compounds (NMVOC), including NOx, released into the atmosphere. Acidification is characterized by the release of H^+, mainly through substances such as SO_2 and NOx, into the environment. For fossil fuel combustion, the high-temperature flame oxidizes nitrogen present in the air, producing large amounts of NOx. In addition, petcoke and hard coal naturally contain sulfur, which is converted into sulfur dioxide SO_2 during combustion. This SO_2 can react with water in the atmosphere to form sulfuric acid. The electricity-related impacts mainly arise from the electricity mix, i.e., power plants that burn fossil fuels to generate electricity. This combustion also releases NOx and SO_2 into the environment. For mining activities, heavy machinery, diesel combustion, and the use of explosives during extraction lead to significant NOx emissions.

Finally, freshwater eutrophication is characterized by the release of P into the environment. The main contributors to this category are electricity production and fossil fuel combustion. Electricity production emits phosphorus mainly through coal-based power generation [6]. Hard coal and petcoke contain traces of phosphorus. When burned, they end up in fly ash that can contaminate the environment.

5.2 Cements sensitivity analysis

In this section, a comparative assessment of five cement types with different material compositions is carried out. These cements mainly differ in their clinker content and in the proportions of supplementary materials, i.e., blast furnace slag, pozzolana, fly ash, and limestone, as shown in Table 1. This variability makes it possible to evaluate how different formulations influence environmental performance.

Environmental impact is strongly driven by clinker content. Since clinker production is the dominant source of environmental burdens, cements with high clinker shares, such as Portland cement, show the highest impacts, while cements with significant clinker substitution, such as Composite cement, exhibit the lowest impacts.

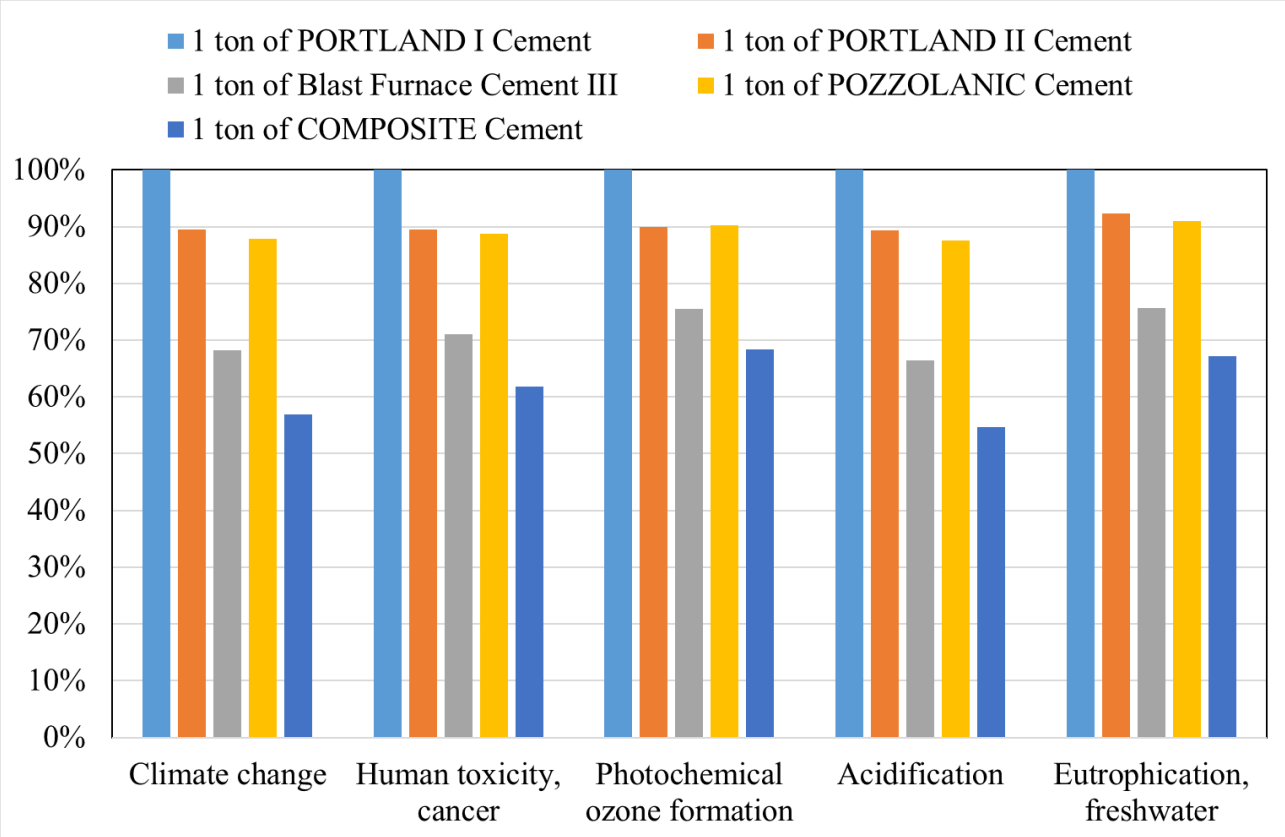


Figure 4: Comparison of environmental impact results for the different cement types

CEM I (Portland Cement) is the reference for the highest impact scenario. Composed of 97.5% clinker, it shows the maximum values across all studied categories. Due to the energy consumption of the clinker process and the chemical emissions from calcination, CEM I is identified as the environmental "worst performer," setting the 100% benchmark for all of the

environmental impacts.

A moderate reduction in environmental impact is observed in **CEM II/A-L(Portland II)** and **CEM IV/A(Pozzonanic)**. By replacing a fraction of the clinker with limestone (13%) or pozzolana and slag (approx. 15%), these cements achieve lower emissions compared to the baseline marked by the CEM I, with reductions of about 11% in every type of impact.

The most significant improvements are found in **CEM III/A (Blast furnace)** and **CEM V/A (Composite)**, which are characterized by using substitution materials like blast furnace slag, pozzolana, and fly ashes. The composite cement delivers the best performance across all of the impact categories with reductions of about 30-45%.

Cements detailed analysis

A detailed comparison of cements is performed to show how the different inventory inputs influence the impacts. The detailed comparison is shown in the next page in Figure 5.

As expected, the Life Cycle Assessment results show that clinker production is the main source of environmental impacts across all categories. However, the analysis also indicates that secondary inputs play a non-negligible role in specific impact categories.

In particular, for CEM IV, the input of blast furnace slag accounts for about 10% of the impact in Human Toxicity with cancer effet and nearly 14% in Acidification. These impacts can be linked to the presence of heavy metals such as chromium, nickel, and lead [7].

Regarding freshwater eutrophication, the results highlight a shift in the relative importance of contributors. As the clinker content decreases in CEM IV and CEM V, the absolute impact of high-voltage electricity remains constant or slightly increases due to grinding requirements, representing about 10–20% of the total impact points for all cement types. This indicates that the relationship between electricity consumption and freshwater eutrophication mainly originates from upstream emissions associated with power generation rather than from the cement production process itself.

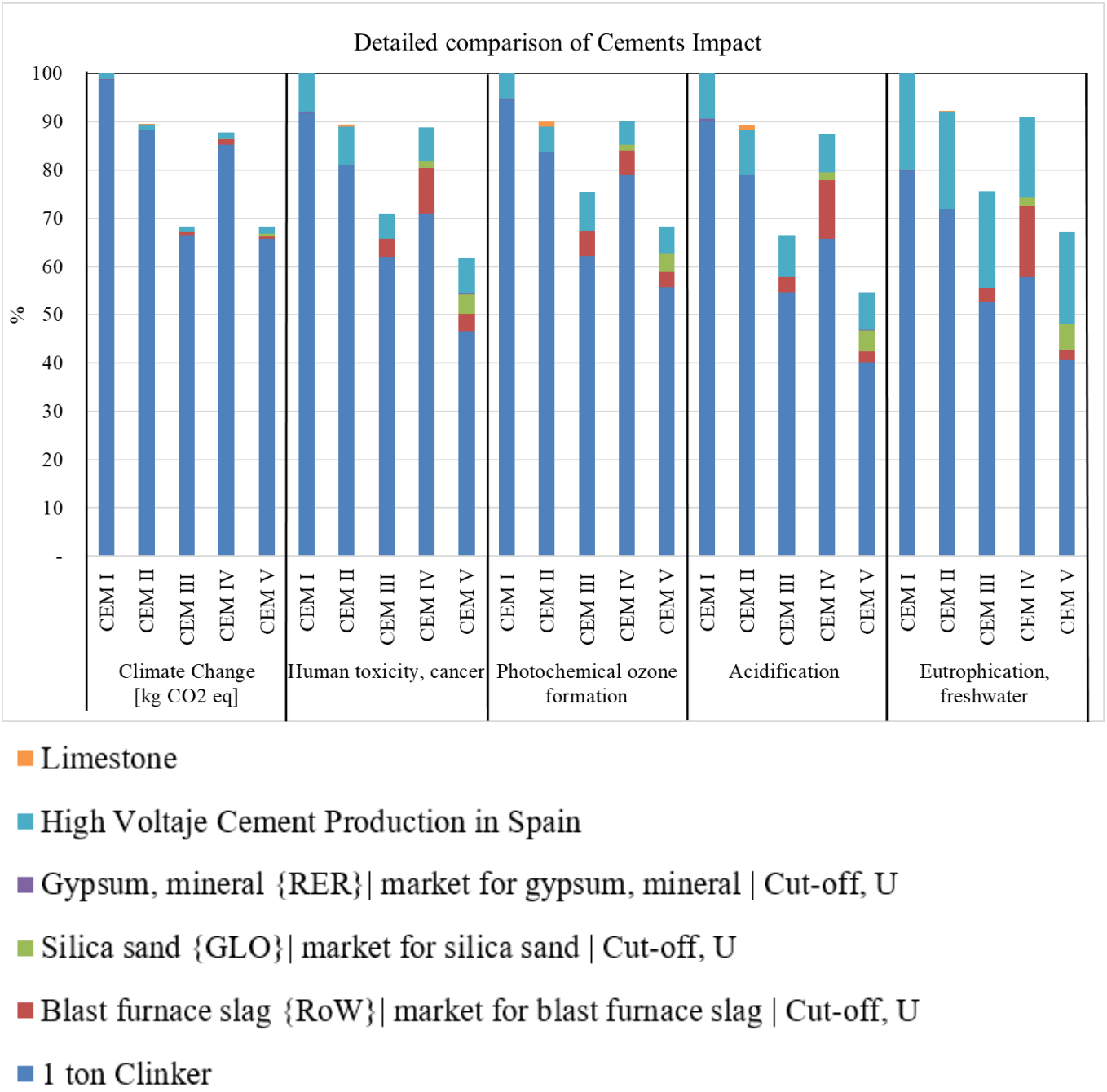


Figure 5: Detailed comparison of Cements Impact

5.3 Critics of the model results for clinker and cement

This LCA is not a complete assessment, as several simplifications were made throughout the modeling process. First, the electricity mix was simplified, with municipal waste modeled as medium-voltage electricity. Second, secondary materials were represented by a single process i.e. blast furnace slag, excluding other materials within this category. Finally, alternative fuels were simplified by selecting the largest inventory contribution, i.e., used tyres.

Based on the results of cement types, it is clear that reducing the clinker content has a direct and significant effect on lowering the environmental impacts of cement production. Switching from normal Portland cement, i.e., CEM I, to blended and more ecological cements, i.e., CEM II, III, and V, is a really viable option, both for the budget and for technical performance. Economically, using less clinker lowers production costs because it saves energy and cuts down on carbon taxes, as shown in reports by Market Growth Reports [8] and the IEA [9]. On the technical side, even though these cements might gain strength a bit slower at the start compared to CEM I, they end up being just as strong in the long run and fully meet the standards [10] [1]. Plus, according to Heidelberg Materials [11], they actually handle chemical attacks better and are less likely to crack, making them a safe and smart choice for most construction projects.

According to the different scenarios, some differences are observed between them, specifically the *Climate Change* and *Freshwater Eutrophication* impacts results can be explained by the assumption that 'used tyres' were the only alternative fuel utilized. Since tyres are fossil-based, they did not provide the expected reduction in carbon emissions; however, this choice unintentionally improved the Eutrophication results by excluding the agricultural waste of typical alternative fuel mixes. It is recognized that including a more diverse range of alternative fuels would have generated better results, closer to the reference scenario.

Allocation choices were also made. The subdivision of clinker production and cement production into two distinct functional units allows a more detailed analysis and a better identification of the critical points within the cement industry.

In addition, this study is limited to a gate-to-gate analysis, meaning that not all life cycle stages are taken into account. The focus is placed on factory operations, i.e., the inputs and outputs of the production system rather than on the full life cycle of the product.

6 Proposed system improvement

This section presents a comparison of clinker production under two different scenarios, where some parameters will be changed: a BASE scenario based on conventional 'fossil fuel' use and an alternative fossil fuel substitution scenario E4 shown in Figure 6. The use of this type of fuel in the cement industry offers the opportunity to reduce production costs, remove of waste and reduce CO2 emissions.

In this study, the new scenario assumes a **50% substitution of fossil fuels by alternative fuels**, while the BASE scenario keeps the original fuel mix proportions reported in the study. The environmental performance of both scenarios is evaluated per 1 ton of clinker.

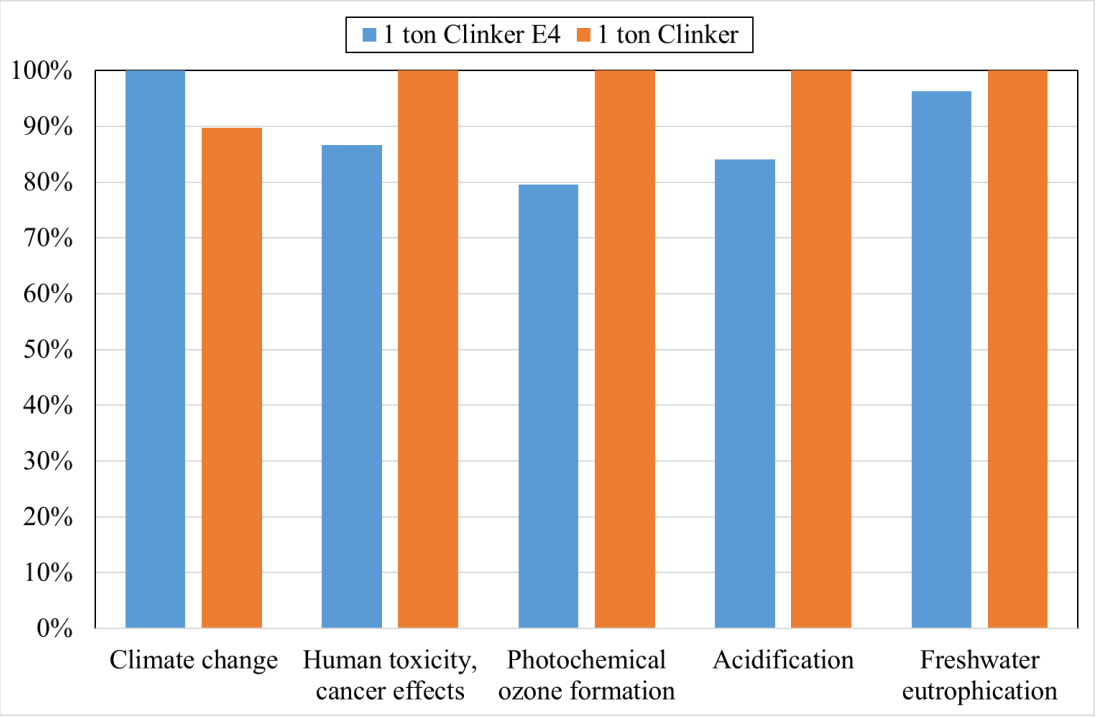


Figure 6: LCA comparison of clinker production under the standard fuel scenario and the E4 fossil fuel substitution scenario.

6.1 Analysis of the scenario

In this analysis, some general improvements can be observed, but they are not as significant as expected. Instead, we find a sort of trade-off between different impact categories.

In case of *Climate Change*, the alternative scenario shows a worsening in the impact, an increase of about 10% compared to the BASE scenario. This result is explained by the nature

of the alternative fuel selected. Since "used tyres" are largely composed of synthetic rubber (derived from petroleum), they are still considered a fossil-based fuel. Their combustion releases fossil CO_2, and when combined with the transport or processing steps required to treat them, the total carbon footprint exceeds that of the fossil fuel mix used in the BASE scenario.

However, for other categories the scenario proves to be better, for example, in *Human Toxicity and Acidification* the model shows a clear reduction of impact. This indicates that the 'Used Tyres' selected from Ecoinvent has a lower sulphur and heavy metal content compared to the 'Hard Coal' used in BASE scenario. In the case of *Freshwater Eutrophication*, a reduction is also observed here, suggesting that tyres do not release significant nitrogen or phosphorus loads compared to fossil fuels.

To improve the environmental performance of the system, the suggested solution is to diversify the fuel mix. That means use more quantity of biomass derived fuels, such as Meat bone and meal, Wood waste, Impregnated sawdust...By adding this mix, the system would benefit from the high energy density from the tyres while simultanoeusly decrease the climate change impact.

6.2 Comparison of the results with the reference report

The results obtained in the proposed LCA model show partial similarities with the results of the reference study. However, significant discrepancies were observed in specific impact categories due to the composition of the alternative fuel mix selected for the simulation.

CC: A notable divergence is observed in this category. In the reference report,the E4 scenario achieves a 92% reduction in emissions compared to the BASE scenario. In contrast, the current LCA model shows a 10% increase in impact for the alternative scenario (E4: 100% vs BASE: 90%). This worse performance of the E4 scenario can be explained by the modeling assumption that 'used tyres' were considered the only alternative fuel. Unlike the reference report, which uses a more diverse mix of alternative fuels (often including biomass or carbon-neutral fractions), used tyres are fossil-based, a feature that affects the *Climate Change* impact results.

HTce: The model is consistent with the reference trend, showing a reduction in impact for the alternative scenario for the *Human toxicity cancer effects* impact. The substitution of coal/petcoke with tyres reduces the release of specific heavy metals associated with coal mining and combustion.

POF: Both the reference report and the LCA model confirm a significant benefit of the alternative scenario. While the reference report shows a higher reduction, the LCA model confirms that alternative fuels contribute less to *Phothochemical ozone formation* compared to the standard fossil fuel mix.

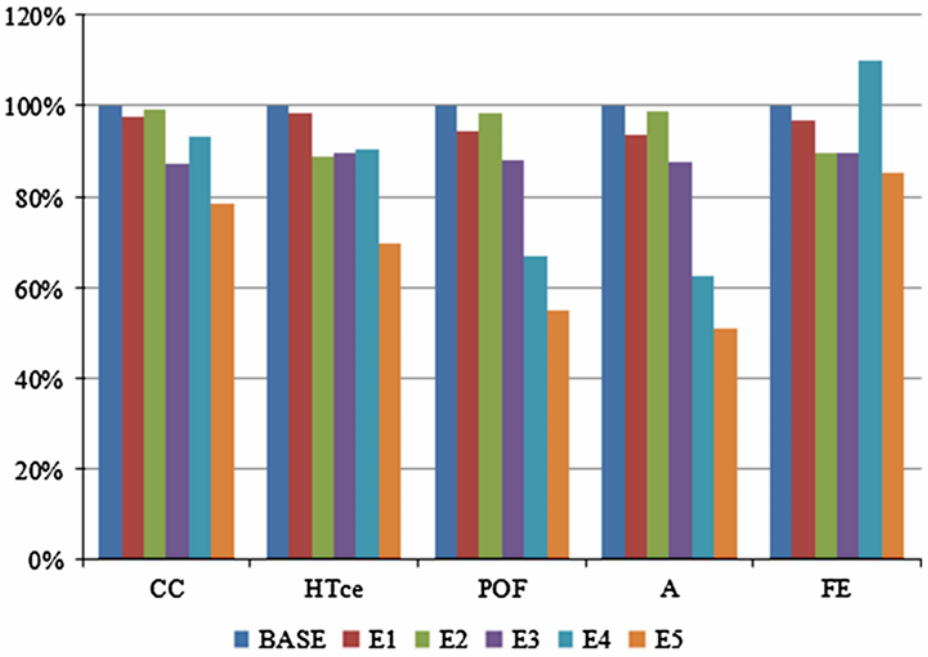


Figure 7: LCA results of the reference report from García et al.

Impact Category	LCA Model		Reference Report	
	E4	BASE	E4	BASE
Climate Change (CC)	100%	90%	92%	100%
Human Tox. Cancer (HTce)	86%	100%	90%	100%
Photochem. Oxidant (POF)	80%	100%	66%	100%
Acidification (A)	83%	100%	62%	100%
Freshwater Eutroph (FE)	96%	100%	110%	100%

Table 7: Comparison of environmental performance percentages between LCA Model and Reference Report.

A: Results are consistent in direction but different in magnitude. The reduction in *Acidification* observed in the LCA model suggests that 'used tyres' have lower sulphur content than the coal they replace, although the reference fuel mix, which includes biomass, achieves even greater reductions.

FE: An inversion of the trend is observed and it has sense. While the reference report indicates that their alternative fuel mix increase eutrophication (associated with bio-waste or organic residues like 'Meat bone and meal' or 'Wood waste'), the LCA model based on 'used tyres' shows a reduction, as tyres do not contain the same levels of nitrogen or phosphorus as other biomass fuels.

7 Conclusion

The LCA study focuses on the Spanish cement industry by analyzing the environmental impacts of clinker production, different cement types, and an alternative fuel scenario using SimaPro and the Ecoinvent database.

The overall results obtained for the base scenario are considered robust and are consistent with the general trends reported in the reference literature. Climate change is identified as the most critical impact category for the Spanish cement industry. Clinker production contributes the largest share of the environmental burden. The results show that CO_2 emissions come mainly from two mechanisms. First being the calcination of limestone and second being the combustion of fossil fuels. Calcination is an unavoidable process that releases CO_2 due to the chemical reaction. But the use of fossil fuels, such as petcoke and coal, can be substituted and reduce impacts on climate change.

The comparison of different cement types confirms that reducing the amount of clinker is an effective way to lower the environmental impacts. Cements with higher proportions of supplementary materials show improved environmental performance, particularly in the climate change category.

The analysis of the alternative fuel scenario shows nuanced results. Although improvements are observed in several impact categories, the climate change impact does not decrease due to the fossil-based nature of used tyres. This shows the importance of the selection alternative fuels by prioritizing biomass or low-carbon options to achieve real reductions in CO_2 emissions.

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8 Appendix

Technology	**kWh (input)**
Hard coal	0.0810
Lignite	0.0040
Natural gas, natural gas combined cycle plant	0.2308
Natural gas, gas turbine	0.0065
Hydropower, large	0.1381
Hydropower, run-of-river	0.0244
Nuclear	0.2215
Solar, photovoltaic	0.0219
Solar, concentrated solar power	0.0025
Wind	0.1544
Biomass	0.0088
Biogas cogeneration	0.0025
Natural gas cogeneration	0.0892
Gas oil, fuel, propane cogeneration	0.0092
Municipal solid waste at incineration plant	0.0052
Electricity from the mix	**kWh (output)**
Electricity	1

Table 8: Electricity mix from Garcìa et al.

Item	Amount
Input	
Primary materials	
Limestone (t)	1.12E+00
Calcarenous marl (t)	7.97E-02
Clay (t)	7.97E-02
Sand (t)	2.73E-02
Iron ore, 46 % Fe (t)	8.58E-03
Kaolin (t)	6.28E-03
Silica sand (t)	3.66E-03
Bauxite (t)	2.61E-03
Feldspar (t)	2.86E-04
Alumina (t)	2.02E-04
Aluminium oxide (t)	1.27E-04
Secondary materials	
Aluminium oxide, waste (t)	3.98E-04
Blast furnace slag (t)	3.16E-03
Carbonized sludges (t)	3.12E-03
Cement, materials (t)	7.19E-04
Clay (recycling) (t)	7.01E-04
Sugar beet limes (t)	9.54E-05
Fly ashes (t)	2.98E-03
Foundry sand (t)	2.74E-04
Iron ore, waste (t)	4.76E-04
Iron recycled (t)	6.59E-03
Mining wastes (t)	1.61E-03
Other secondary materials (t)	1.86E-03
Other slag from meal (t)	4.45E-04
Pyrite ashes (t)	2.92E-03
Industrial solids (t)	1.00E-03
Water (m^3)	1.62E-03
Infrastructure	
Industrial machine, heavy (t)	3.76E-05
Cement plant (p)	6.27E-12
Transport	
Conveyor belt (km)	2.00E+00
Lorry 20-28 t (tkm)	3.19E+00
Lorry >28 t (tkm)	1.89E+01
Fossil fuels	
Petroleum coke (GJ)	2.89E+00
Heavy fuel oil (GJ)	3.41E-02
Natural gas (GJ)	4.36E-03
Diesel (GJ)	5.72E-04
Hard coal (GJ)	4.37E-02
Alternative fuels	
Other liquid fuels biomass (GJ)	3.23E-05
Used tyres (GJ)	1.72E-01
Meat bone and meal (GJ)	4.96E-02
Municipal sewage sludge (GJ)	3.01E-02
Refuse-derived fuel MSW (GJ)	9.73E-02
Wood waste (GJ)	5.76E-03
Impregnated sawdust (GJ)	4.17E-02
Varnishes and solvents (GJ)	4.54E-02
Used oils (GJ)	6.57E-03
Plastics (GJ)	2.53E-02
Pulp, paper, paperboard (GJ)	5.80E-04
Textile waste (GJ)	3.14E-04
Others no biomass (GJ)	3.15E-02
Hydrocarbon residues (GJ)	3.65E-03
Electricity	
Electricity (MWh)	9.20E-02
Output	
Emissions	
Carbon dioxide (process) (t)	5.28E-01
Particulates (PM_{10}) (t)	1.04E-05
Products	
Clinker (t)	1.00E+00

Table 9: Clinker Production Data from Garcìa et al.

		Petcoke	Hard coal	Heavy fuel oil
Input				
Input	kg	1	1	1
LHV	MJ/kg	32.21	26.26	40.19
Output				
Emissions to air				
Carbon dioxide, fossil	kg	3.19E+00	2.26E+00	3.09E+00
Carbon monoxide, fossil	kg	3.12E-03	–	4.90E-03
NMVOC	kg	4.83E-05	–	1.21E-03
Nitrogen oxides	kg	6.44E-03	1.49E-02	7.78E-03
Sulphur dioxide	kg	2.19E-02	4.18E-03	3.07E-02
Benzene	kg	6.50E-07	6.50E-07	2.67E-08

Table 10: Input/Output and Emissions Data for Different Fuels

NMVOC refers to non-methane organic volatile compound

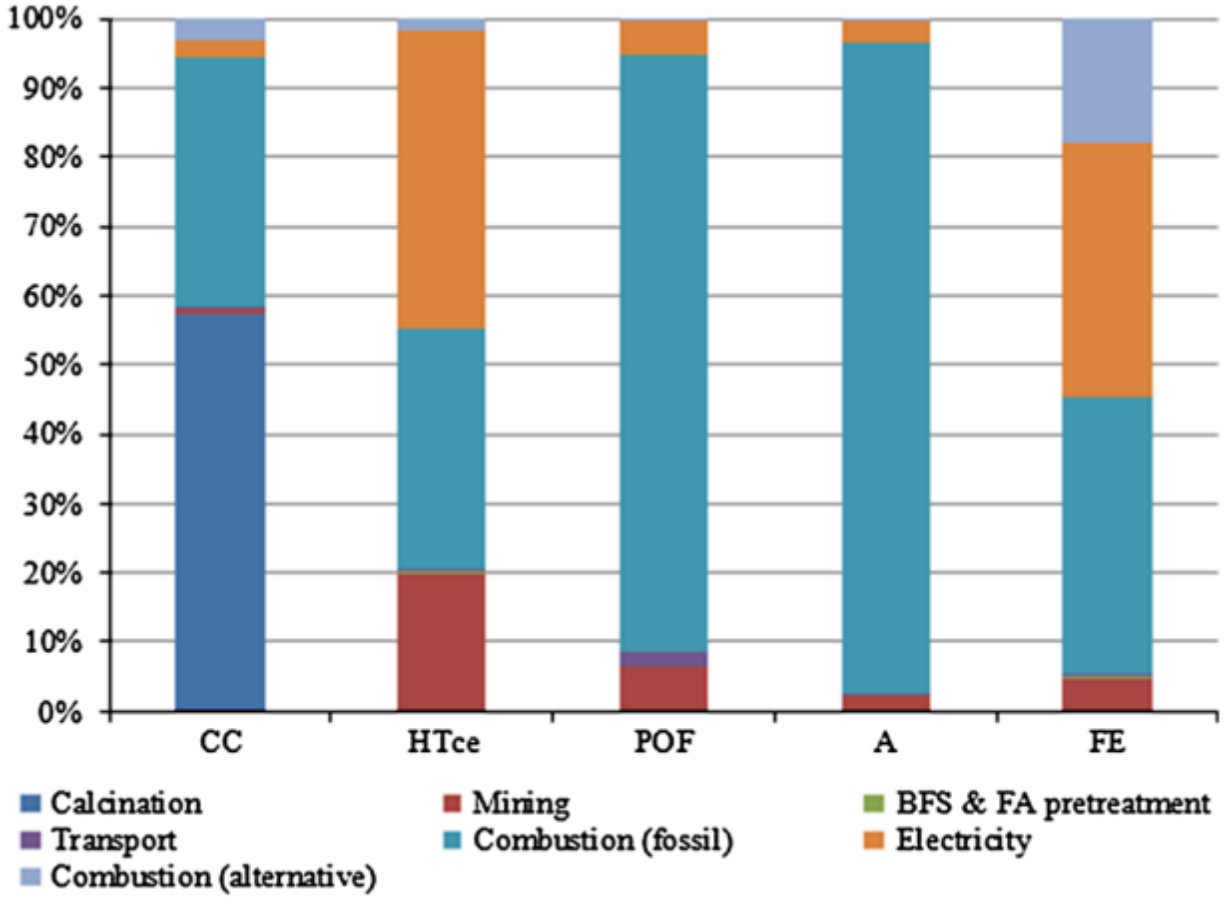


Figure 8: LCIA results of the production of 1 t of clinker in 2010 by production stage Garcìa